Straight exercises from courses 1-3

Exercice : State-Space Manipulation

Let us consider the system defined by the following block diagram

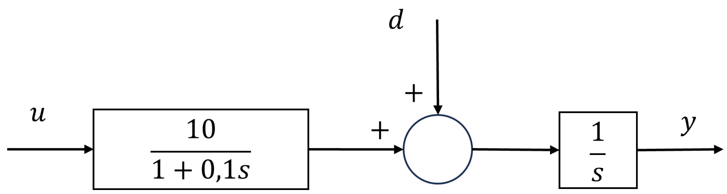

Figure 1 – Block Diagram of the studied system

1. Give the state-space representation, considering u and d as inputs, and y as the output.
2. Study the controllability with respect to u

We now suppose that the input disturbance d is constant,

3. Define an augmented state which includes this variable as an internal variable. What is the state space representation with this augmented state space ?
4. Study its controllability, its observability
5. Proceed to the controllability canonical decomposition

Exercice : Linearization model of a food network

Consider the mathematical model of a food network (set of food chains) given by the following system :

$$\left\{ \begin{array}{lcl} \dot{x}_1 & = & -x_1 + x_1 x_2 \\ \dot{x}_2 & = & -x_2 - x_1 x_2 + x_2 x_3 \\ \dot{x}_3 & = & -x_3 - x_2 x_3 + u \end{array} \right.$$

where x_1 denotes the top predator (e.g. killer whale), x_2 denotes the intermediate predator/prey (e.g. seal), x_3 denotes the final prey (e.g. a phytophagous fish) and u refers to the food (e.g. algae). We are interested in the top predator evolution.

1. Given a constant food quantity u_0, what are the different analytical expressions of the equilibrium points ?
2. Give the expression of the linearized models around with equilibrium points.
3. We suppose now that $u_0 = 3$. Study the stability and the controllability of the linearized models.
4. What do you think on the ability to ensure that the top predator population remains constant ?